

B. Tech Degree VI Semester Examination April 2011**ME 604 HEAT AND MASS TRANSFER**
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A
(Answer ALL questions)

(8 x 5 = 40)

- I. (a) What is critical insulation thickness? Derive an expression for the critical insulation thickness for a cylinder.
- (b) Derive an expression for the temperature distribution in a semi infinite plate.
- (c) Differentiate film boiling and pool boiling.
- (d) State and explain Newton's law of cooling. Differentiate Free Convection and Forced Convection. Enumerate the differences between Laminar flow and Turbulant flow.
- (e) What do you mean by Radiation Shape factor? State and explain reciprocity theorem.
- (f) Describe about Radiation Shields. How do you find out the percentage reduction in heat flow by using radiation shields?
- (g) Explain about different types of Heat Exchangers.
- (h) Describe about
- Effectiveness
 - NTU
 - LMTD
 - Fouling factor

PART – B

(4 x 15 = 60)

- II. A steel tube ($K = 43.26 \text{ W/mK}$) of 5.08 cm I.D and 7.62 cm O.D is covered with a 2.54 cm layer of asbestos insulation ($K = 0.208 \text{ W/mK}$). The inside surface of the tube receives heat by convection from a hot gas at a temperature of $T_a = 316^\circ\text{C}$ with a heat transfer co-efficient $h_a = 284 \text{ W/m}^2\text{K}$ while the outer surface of the Insulation is exposed to the ambient air at $T_b = 38^\circ\text{C}$ with a heat transfer coefficient of $h_b = 17 \text{ W/m}^2\text{K}$. Estimate
- the loss of ambient air for 3m length of the tube
 - the temperature drops across the tube material and insulation layer.

(15)

OR

- III. (a) A hollow sphere is kept at a uniform temperature T_i at the inner surface $r = r_i$ and at temperature T_o at the outer surface $r = r_o$. Show that the heat transfer rate, Q , through the sphere is given by

$$Q = \frac{T_i - T_o}{R} \text{ where the thermal resistance of the sphere is } R = \frac{r_o - r_i}{KA_m},$$

$$A_m = \sqrt{A_i A_o}.$$

(7)

- (b) A hollow sphere is made up of two materials; first with $K = 70 \text{ W/mK}$ is having an I.D of 10cm and O.D of 30cm and the second with $K = 15 \text{ W/mK}$ forms the outer layer with O.D of 40cm. The inside and outside temperatures are 300°C and 30°C respectively. Estimate the rate of heat flow through this sphere assuming perfect contact between two materials.

(8)

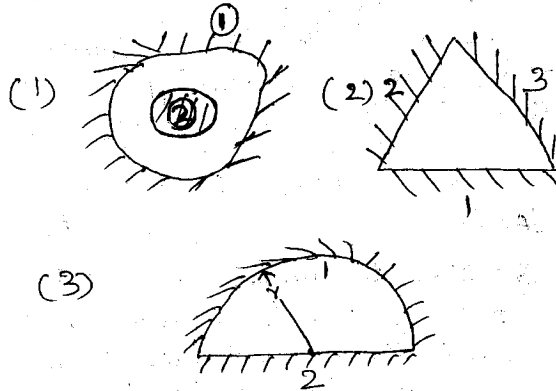
(P.T.O)

- IV. A large vertical plate 5m high is maintained at 100°C and exposed to air at 20°C . Calculate the convection heat transfer coefficient. (15)

OR

- V. Atmospheric air at 275 K and a free stream velocity of 20m/s flows over a flat plate of 1.5m long that it maintained at a uniform temperature of 325 K. Calculate the average heat transfer co-efficient over the region where the boundary layer is laminar, the average heat transfer co-efficient over the entire length of the plate and the total heat transfer rate from the plate to the air over the length 1.5m and width 1m. Assume transition occurs at $Re_c = 2 \times 10^5$. (15)

- VI. (a) Compute the shape factors between different surfaces of the configurations shown in the figure.



- (b) A small sphere (O.D = 60mm) with a surface temperature of 300°C is located at the geometric centre of a large sphere (I.D = 360 mm) with an inner surface temperature of 15°C . What is the net interchange of heat between the two spheres? (8)

OR

- VII. The net radiation from the surface of two parallel plates maintained at temperatures T_1 and T_2 is to be reduced by 79 times. Calculate the number of screens to be place between the two surfaces to achieve this reduction in heat exchange, assuming the emissivity of the screens as 0.05 and that of the surfaces as 0.8. (15)

- VIII. A counter flow heat exchanger is employed to cool 0.55 Kg/s ($C = 2.45$ kJ/Kg $^{\circ}\text{K}$) of oil from 115°C to 40°C by the use of water. The inlet and outlet temperatures of cooling water are 15°C and 75°C respectively. The over all heat transfer co-efficient is expected to be 1450 W/ m^2K . Using NTU method calculate the following.

- (i) The mass flow rate of water
- (ii) The effectiveness of the heat exchanger
- (iii) The surface area required. (15)

OR

- IX. In a certain double pipe heat exchanger, hot water flows at a rate of 5000 Kg/hr. and gets cooled from 95°C to 65°C . At the same time, 50,000 kg/hr of cooling water at 30°C enters the heat exchanger. The over all heat transfer coefficient remains constant at 2270W/ m^2K . Determine the heat transfer area required and the effectiveness assuming that two streams are in parallel flow. Assume for the both the streams $C_p = 4.2\text{kJ/KgK}$. (15)